



Javier Salazar Cavazos

📍 Ann Arbor, MI, USA ✉ javiersc@umich.edu 📧 javiersc.phd
 🌐 javiersalazar6 🎓 Google Scholar 📄 javiersc1 📞 0009-0009-1218-9836

Education

- PhD** 📄 **University of Michigan**, Electrical & Computer Engineering Ann Arbor, MI, USA
Aug 2021 – May 2026
- Dissertation: [Learning Representations from Noisy Data and Brain Imaging](#) 📄
 - Research Topics: Heteroscedastic Data, Subspace Modeling, Deep Clustering, Behavior Score Prediction, and Optimization Algorithms for Inverse Problems
 - Advisors: [Laura Balzano](#) 📄, [Jeffrey Fessler](#) 📄, and [Scott Peltier](#) 📄
- MS** 📄 **University of Michigan**, Electrical & Computer Engineering Ann Arbor, MI, USA
Aug 2021 – May 2023
- GPA: [3.95/4.0](#) 📄, with highest distinction
 - Concentration: Signal/Image Processing & AI/Machine Learning
- BS** 📄 **University of Texas**, Electrical Engineering Arlington, TX, USA
Aug 2015 – May 2020
- GPA: [4.0/4.0](#) 📄, summa cum laude
 - Capstone Project: [Wheelchair Dynamometer](#) 📄
- BS** 📄 **University of Texas**, Mathematics Arlington, TX, USA
Aug 2015 – May 2020
- GPA: [4.0/4.0](#) 📄, summa cum laude

Industry Experience

- KLA** 📄, Algorithms Engineer Ann Arbor, MI, USA
June 2026 – present
- Creating machine learning and image/signal processing algorithms for semiconductor defect inspection to support fab foundries in process node developments.
- KLA** 📄, Algorithm Engineering Intern Ann Arbor, MI, USA
May 2025 – Aug 2025
- Worked with the reticle/photomask electron beam team on an image processing problem to mitigate artifacts present in SEM images for better defect inspection.
 - Designed a fast algorithm to remove artifact structures that made it into the production codebase for real-world use. [Click here for image](#) 📄.
- KLA** 📄, Algorithm Engineering Intern Ann Arbor, MI, USA
May 2024 – Aug 2024
- Worked with the reticle/photomask team to group majority image patterns and outliers to augment defect classification systems.
 - Designed deep learning clustering and outlier detection algorithms to reduce customer intervention time in reticle development. [Click here for image](#) 📄.
- Precision Design** 📄, Co-Founder Arlington, TX, USA
2020 – present
- I designed an aftermarket 3rd brake light for Ford F150/Raptor model trucks. Myself, along with my business partner, sell these products on our site [precisiondesign.us](#) 📄. We have shipped over 500 units to date. [Click here for product drawings](#) 📄.
- Olsson** 📄, Assistant Electrical Engineer Fort Worth, TX, USA
Aug 2020 – Aug 2021
- Designed electrical power distribution systems for data center clients and analyzed systems for short circuit analysis, arc flash mitigation, and breaker setting coordination. [Click here for a time current curve \(TCC\) example](#) 📄.
- Brohard Architecture** 📄, Architectural Draftsman Dallas, TX, USA
Sept 2017 – July 2019

- Designed building models using Autodesk Revit and created floor plans, ceiling plans, interior/exterior elevation plans, and perspective plans for commercial & residential projects. [Click here for an example project](#).

Premier Surveying, AutoCAD Technician

Plano, TX, USA
Jan 2015 – Aug 2017

- Drafted residential title surveys using Autodesk AutoCAD Civil3D and analyzed field data, city plats, FEMA flood maps, title commitments, and easements to construct drawings. [Click here for a title survey example](#).

Publications

Behavior Score Prediction in Resting-State Functional MRI by State Space Modeling <i>Javier Salazar Cavazos</i> , Maximillian Egan, Krisanne Litinas, Benjamin Hampstead, Scott Peltier 10.48550/arXiv.2602.07131 (IEEE Journal of Biomedical and Health Informatics)	2026
Smooth Optimization Using Global and Local Low-Rank Regularizers Rodrigo Lobos, <i>Javier Salazar Cavazos</i> , Raj Rao Nadakuditi, Jeffrey A. Fessler 10.1137/25M175891X (SIAM Journal on Imaging Sciences)	May 2026
PET-TURTLE: Deep Unsupervised Support Vector Machines for Imbalanced Clusters <i>Javier Salazar Cavazos</i> 10.1109/LSP.2025.3636453 (IEEE Signal Processing Letters)	Nov 2025
ALPCAHUS: Subspace Clustering for Sample-wise Heteroscedastic Data <i>Javier Salazar Cavazos</i> , Jeffrey A. Fessler, Laura Balzano 10.48550/arXiv.2505.18918 (IEEE Transactions on Signal Processing)	2025
Alzheimers Disease Classification in Functional MRI With 4D Joint Temporal-Spatial Kernels in Novel 4D CNN Model <i>Javier Salazar Cavazos</i> , Scott Peltier 10.48550/arXiv.2506.02060 (International Society for Magnetic Resonance in Medicine)	May 2025
ALPCAH: Subspace Learning for Sample-wise Heteroscedastic Data <i>Javier Salazar Cavazos</i> , Jeffrey A. Fessler, Laura Balzano 10.1109/TSP.2025.3537867 (IEEE Transactions on Signal Processing)	Jan 2025
Sample-wise Heteroscedastic PCA with Tail Singular Value Regularization <i>Javier Salazar Cavazos</i> , Jeffrey A. Fessler, Laura Balzano 10.1109/SampTA59647.2023.10301206 (Sampling Theory and Applications)	Aug 2023

Certifications

Linux Essentials, **Linux Professional Institute (LPI)**: Covered the fundamentals such as basic concepts of open source, Linux command line, understanding files and scripts, and security/permissions

A+, **Computing Technology Industry Association (CompTIA)**: Covered various aspects of computer hardware, software, networking, security, and troubleshooting to manage IT infrastructures

Solidworks Associate in Mechanical Design, **Dassault Systèmes**: Covered the fundamentals of 3D modeling, assembly design, and simulation using Solidworks software

AutoCAD Professional, **Autodesk**: Covered the fundamentals of 2D drafting and design documentation using AutoCAD software

F.E. Electrical and Computer, **National Council of Examiners for Engineering and Surveying (NCEES)**: Covers topics such as circuits, electronics, control systems, and digital systems

Languages

Programming: Proficient with Python, Pytorch, Julia, LaTeX; basic familiarity with MATLAB and C

Traditional: English (native), and Spanish (native)

Awards

ISMRM 2026 Best Paper: The ISMRM 2026 conference gives a best paper award (summa cum laude) for the top 5% highest scoring submissions showing high interest, quality, and impact regarding our research submission on behavior score prediction in resting-state functional MRI

Michigander Scholars in Semiconductors Fellowship: The Michigander Scholars Program, funded by the Michigan Economic Development Corporation (MEDC) and the TAT (Talent Action Team), aims to enhance Michigan's talent pool in the semiconductor sector. The University of Michigan offers this program to students pursuing degrees in computer, electrical, process, or software engineering. The main goal of this program is to support students who are pursuing careers in the broader semiconductor domain, and are planning to launch their career within the Michigan state

Rackham Merit Fellow (RMF): The RMF Award is a highly competitive named fellowship awarded to students who have outstanding academic qualifications, show exceptional potential for scholarly success in their graduate program, and demonstrate promise for contributing to wider academic, professional, or civic communities

Mathematic Academic Excellence (MAE): The MAE Award is a departmental award granted by the math department from the University of Texas system to students that demonstrate potential and excellence in the field of mathematics

Projects

Proximal Optimized Gradient Method (POGM) [↗](#)

- The POGM optimization algorithm implemented in Python for gradient descent applications, adapted from Jeff Fessler Julia code. See original paper for more details. Code located at <https://github.com/javiersc1/ProximalOptimizedGradientMethod>.

Image Patch Extraction [↗](#)

- Python code to extract patches from an image, do some operation, and unpatchify it. Supports overlapping patches. Code located at <https://github.com/javiersc1/Patchify>.

Modern Vision architectures [↗](#)

- Implemented various modern vision architectures such as Vision Transformer, ConvNeXT, MLP-Mixer, and VAE, for Alzheimers disease classification in structural MRI using the ADNI dataset.

Human Pose Estimation [↗](#)

- Implemented a vision transformer (ViT) model for human pose estimation using the COCO dataset. The model predicts the locations of keypoints on the human body from images.

Matrix Factorization Review [↗](#)

- Wrote a literature review paper on solving matrix factorization problems that are useful for things like matrix sensing, phase retrieval, and PCA-like problems.

Metropolis-Hastings Sampling [↗](#)

- Implemented the Metropolis-Hastings (Markov chain Monte Carlo) algorithm for sampling from complex distributions. This method is widely used in Bayesian statistics for generating samples from posterior distributions.

Zeroth-Order Optimization Methods [↗](#)

- Implemented various zeroth-order optimization algorithms (no gradient function, only cost function). These methods are useful for optimizing black-box functions where gradient information is not available or expensive to compute.

Computer Vision Algorithms [↗](#)

- Implemented various classical and deep learning methods for computer vision tasks in Python/PyTorch such as: scene recognition (VGG network), panorama stitching (with SIFT features), image blending (laplacian pyramids), image generation (diffusion models), image autoencoders (VAE), Neural style transfer (GANs), and object detection (FCOS detector w/ CNN).

Wheelchair Dynamometer [↗](#)

- Designed and built the software and electrical hardware components of a wheelchair dynamometer to measure the power output of wheelchair athletes. This system acts as a treadmill for wheelchair users.

Machine Learning Algorithms [↗](#)

- Implemented various machine learning algorithms from scratch in MATLAB such as linear regression (LR), linear discriminant analysis (LDA), expectation maximization (EM), k-Means clustering, k-Nearest Neighbors (kNN), and multilayer perceptron (MLP).

Stochastic Models & Simulations [↗](#)

- Stochastic modeling is useful for many real-world scenarios. From random number generation, Markov chains, random walks, and Black-Scholes option pricing, and more. I implemented simulations of these processes in Mathematica to analyze their properties and behaviors.

Numerical Analysis Algorithms [↗](#)

- Numerical analysis is the study of mathematical numerical algorithms such as ODE solvers, Taylor approximations, matrix solvers, root finders, numerical integration, and more. In this project, I implemented several numerical analysis algorithms in MATLAB from scratch and analyzed their performance and accuracy.

Image/Video Compression [↗](#)

- Implemented two different compression techniques DPCM and DCT for video compression. The DCT method is commonly used in MPEG/JPEG algorithms for MP4/JPG files.

WiFi Antenna Design [↗](#)

- Designed, optimized, and tested a 2.4GHz/5GHz wifi antenna using ANSYS HFSS modeling software and electronic equipment such as a vector network analyzer (VNA) and spectrum analyzer.

Teaching

EECS551, Matrix Methods for Signal Processing, Data Analysis, & Machine Learning, University of Michigan: This is a course where I lead a weekly discussion that covers coding algorithms in Julia for machine learning tasks using linear algebra techniques. Moreover, we go through example questions to reinforce course material and exam preparation. This is besides the usual GSI expectations of dealing with homework assignments, tests, and office hours

EECS351, Digital Signal Processing, University of Michigan: This is a DSP course where I lead a weekly discussion reinforcing course material through further mathematical analysis and examples

Undergraduate Research Opportunity Programs

MRI Mesh & Sling Segmentation [↗](#), Research Assistant, UT Southwestern Medical Center

Dallas, TX, USA
Aug 2018 – Sept 2019

- Researched machine learning methods of identification and segmentation of meshes & slings, along with common pelvic floor features, in MR images for radiologist assistance in 3D visualization-guided surgery planning.
- Presented [Automatic Segmentation & 3D Visualization of MR Images of Pelvic Floor](#) [↗](#) at the [Modern Challenges in Imaging](#) [↗](#) conference located at Tufts University, Medford, Massachusetts.

Iceberg Path Prediction [↗](#), Summer Intern, Dartmouth College

Hanover, NH, USA
May 2018 – Aug 2018

- Researched the mathematics of Bayesian inference methods for iceberg prediction tracking by modeling partial differential equations, simulated the model using Markov chain Monte Carlo sampling techniques, and predicted drag coefficients for iceberg path modeling.

Mathematics of Medical Imaging [↗](#), Research Intern, UT Arlington

Arlington, TX, USA
Aug 2016 – Aug 2017

- Researched and implemented the mathematics of computerized tomography such as Radon transform and filtered backprojection for image reconstruction of artificial phantoms.